

머신러닝 프로그래밍

Machine Learning Programming

Lecture 13 : 이상감지 관련 논문



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Temporal-Frequency Masked Autoencoders for Time Series Anomaly Detection

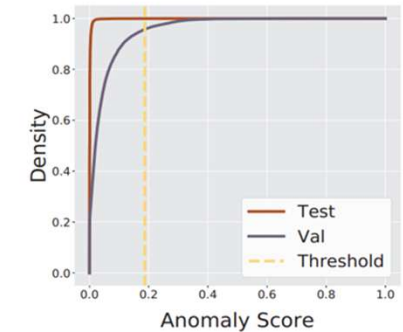
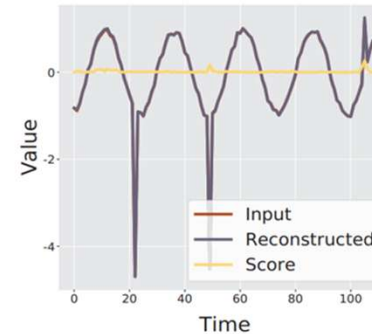
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Introduction

셀프 슈퍼바이저드 러닝

Motivation: The Limits of Current SSL

- ▶ **Labeled Data Scarcity:** Real-world industrial data rarely has comprehensive fault labels.
- ▶ **Limitations of Contrastive Learning (CL):**
 - ✓ **Abnormal Bias:** If unlabeled training data contains hidden anomalies, CL models mistakenly learn them as 'normal' patterns.
 - ✓ **False Alarms:** Minor environmental shifts are often misclassified as anomalies.
- ▶ **Why Masked Autoencoders (MAE)?** By focusing on reconstructing masked inputs rather than comparing samples, MAEs learn the intrinsic normal prototype, showing strong immunity to abnormal bias.

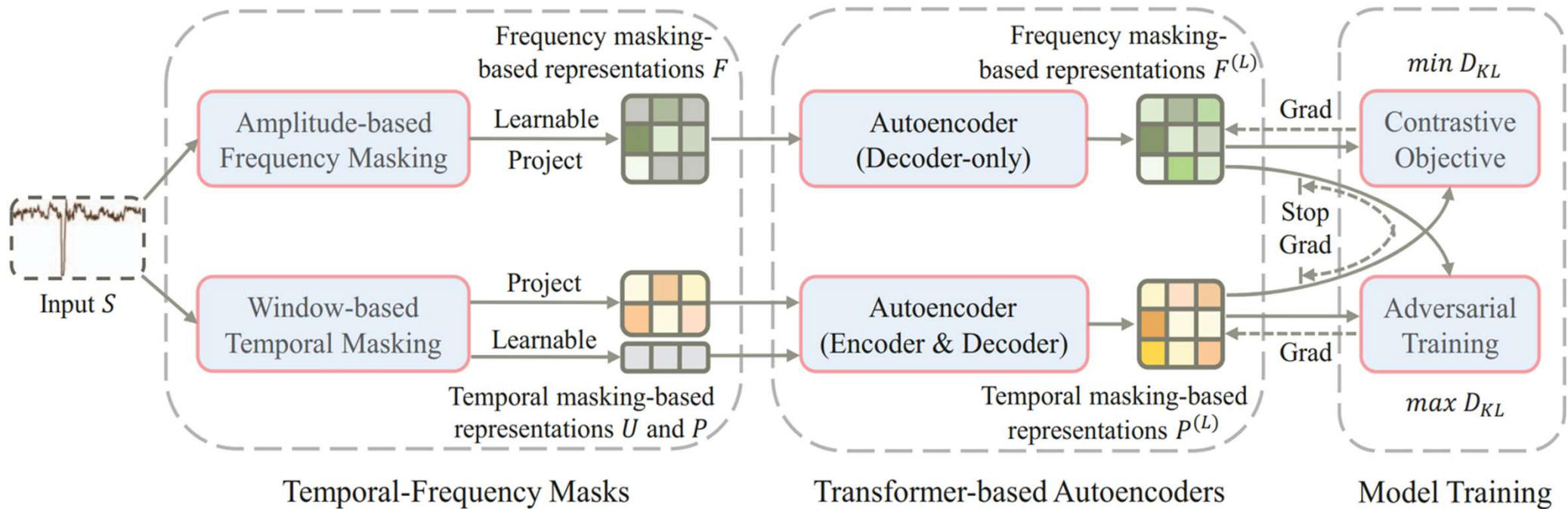


TFMAE Architecture

Problem Definition: The Missing Piece

- ▶ Existing MAEs in TSAD focus solely on the **Temporal domain**.
- ▶ However, 1D time-series data carries crucial physical properties in its **Frequency domain** (e.g., periodic vibrations, spectral amplitudes).
- ▶ Temporal-only reconstruction often fails to capture subtle, frequency-level anomalies.
- ▶ **Goal:** To develop a Dual-Branch MAE that captures both temporal trends and spectral periodicities.

TFMAE Architecture



Dual-Branch Masked Autoencoder

- ▶ The TFMAE framework processes data through two parallel branches:
- ▶ **Temporal Branch:** Masks and reconstructs temporal patches.
- ▶ **Frequency Branch:** Converts to the frequency domain via FFT, then masks and reconstructs amplitude components.
- ▶ **Cross-Domain Alignment:** Ensures representations from both domains are consistent.

Method: Masking Strategies

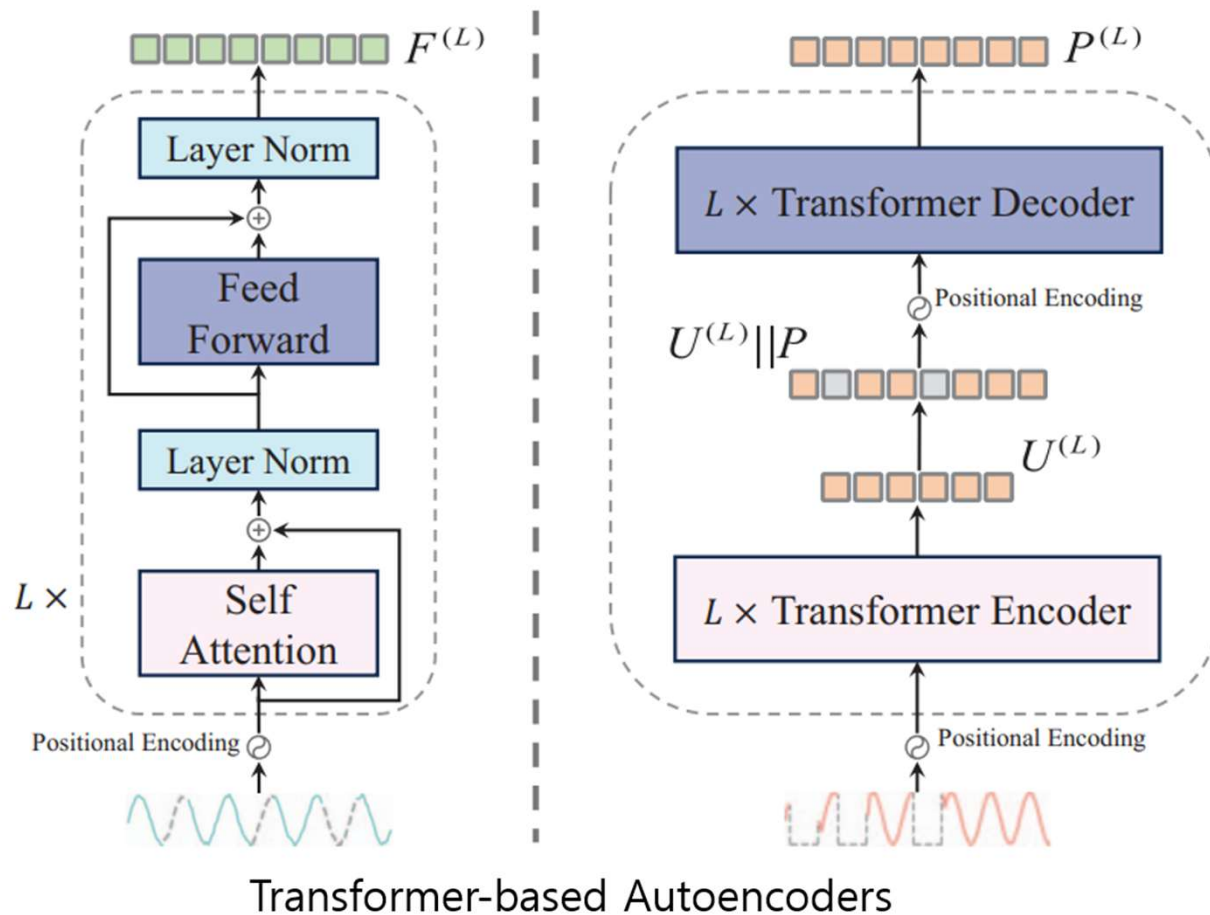


Overview of window-based temporal masking

Overview of amplitude-based frequency masking

- ▶ **Temporal Masking:** Divides the signal into non-overlapping patches.
 - ✓ **Masks patches with larger variations** to eliminate 75% of observed anomalies.
- ▶ **Frequency Masking:**
 - ✓ Computes the amplitude spectrum using Fast Fourier Transform (FFT).
 - ✓ **Masks components with smaller amplitudes** to remove pattern anomalies.
- ▶ **Efficiency:** Only the unmasked ('visible') patches/components are fed into the Transformer encoders.

Method: Transformer-based Autoencoders



- Left: Transformer-based decoder-only autoencoder for recovering masked frequencies, where $\oplus$ indicates elementwise addition.
- Right: Transformer-based autoencoder for recovering masked observations, where $||$ denotes inserting masked representations into learned unmasked representations.

Method: Model Training

▶ **Joint Training Loss:**

- ✓ **Reconstruction Loss:** Mean Squared Error (MSE) between original and reconstructed data in both domains.
- ✓ **Contrastive Loss:** Minimizes the discrepancy between Temporal and Frequency embeddings to align cross-domain knowledge.

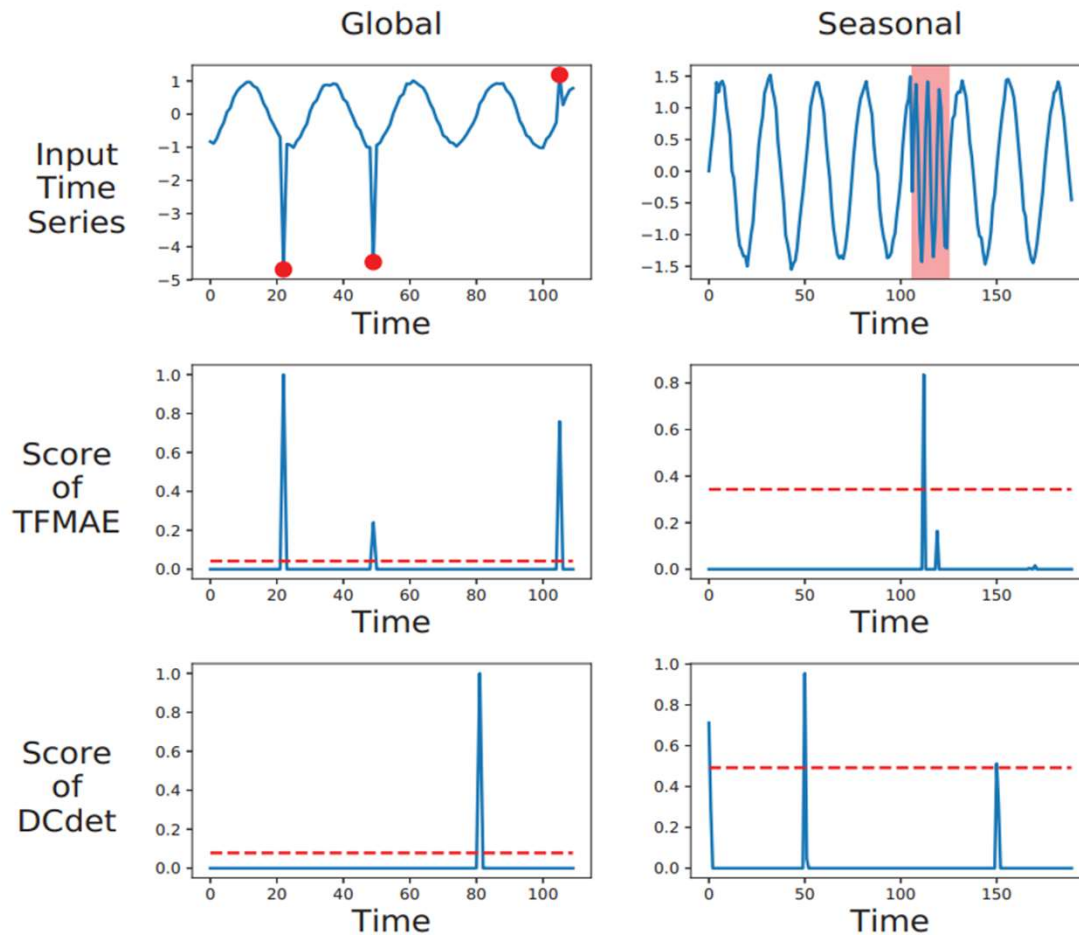
▶ **Anomaly Scoring (Inference):**

- ✓ The model attempts to reconstruct incoming data as 'normal'.
- ✓ **Anomaly Score = Temporal Recon Error + Frequency Recon Error.**
(Higher error = Anomaly).

Experiment

- ▶ **Datasets & Baselines**
- ▶ **5 Benchmark Datasets:**
 - ✓ **SMD** (Server Machine Dataset)
 - ✓ **MSL / SMAP** (NASA Spacecraft Telemetry)
 - ✓ **SWAT / WADI** (Water Treatment & Distribution Systems)
- ▶ **Baselines:** Compared against 10+ SOTA models including Anomaly Transformer, DCdetector, TimesNet, and TranAD.
- ▶ **Evaluation Metrics:** Precision, Recall, and F1-Score.

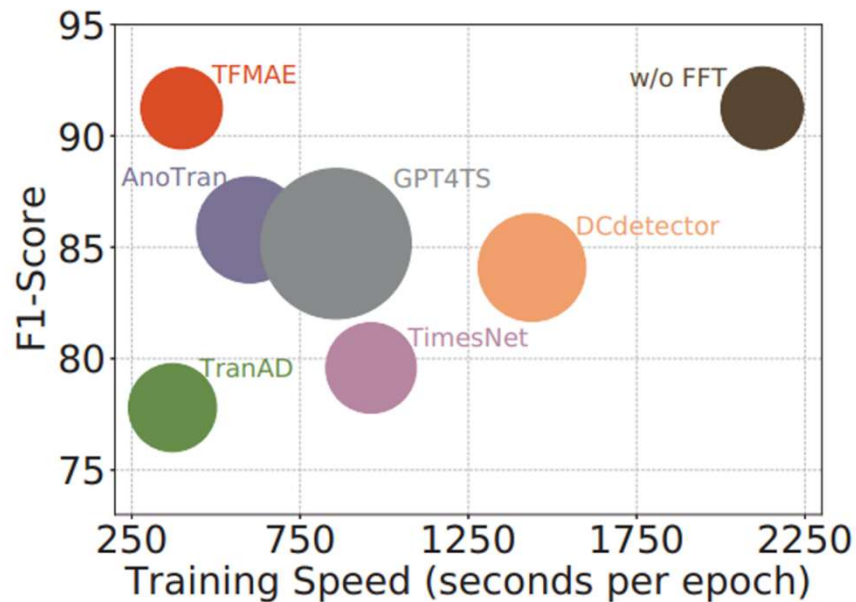
Experimental Results



Robustness to Abnormal Bias

- ▶ **Experiment:** Training data was intentionally contaminated with a high ratio of anomalies (up to 10%).
- ▶ **Result:** While Contrastive Learning models (like Anomaly Transformer) suffered severe performance drops due to abnormal bias, **TFMAE maintained stable accuracy.**
- ▶ **Reason:** The masked reconstruction task forces the model to learn only the dominant, normal physical structure.

Experimental Results



Qualitative & Computational Analysis

- **Reconstruction Plot:** Visualizations show TFMAE successfully reconstructs normal patterns but completely fails to reconstruct anomalous spikes (which is ideal for AD).
- **Feature Distribution (t-SNE):** Normal and anomalous features are distinctly separated in the latent space.
- **Efficiency:** Processing only the unmasked tokens (25% of data) makes TFMAE computationally lighter and faster to train than full-sequence Transformers.

Conclusion

- **A Novel Paradigm:** Proposed TFMAE, successfully integrating Time and Frequency domain representations within a Masked Autoencoder framework.
- **Solving Abnormal Bias:** Effectively mitigated the abnormal bias and distribution shift problems prevalent in contrastive learning-based AD models.
- **SOTA Performance:** Achieved state-of-the-art anomaly detection accuracy and high robustness across multiple real-world industrial datasets.
- **Efficiency:** Demonstrated that high-ratio masking significantly reduces computational overhead while actually improving representational learning.